



OPEN Music and body motion contribute asymmetrically to emotion perception in traditional Chinese plucked-instrument performance

Kaiyuan Ma¹, Biyun Zhang^{1,2} & Jun He¹✉

Expressive body motion is an important but difficult-to-interpret channel for emotion communication in embodied agents. Using musical performance as a naturalistic multimodal context, we examined how audio and body motion contribute to emotion perception. In a controlled perceptual study, the audiovisual percept was more similar to the audio-only percept than to the visual-only percept, consistent with an audio-dominant pattern of integration, while body motion alone yielded weak and ambiguous judgments, particularly for valence. We found no evidence that audiovisual presentation systematically shifted mean valence or arousal, and the overall emotional-intensity effect was sensitive to statistical assumptions. The clearest robust pattern was that valence judgments were least decisive in the visual-only condition, while exploratory clip-level analyses suggested a possible facilitatory effect of body motion for low-arousal clips. To support this analysis, we present EMOSIC, a synchronized dataset of over 7.5 h of monophonic audio and upper-body motion capture from professional performances on traditional Chinese plucked instruments, which provides a non-Eurocentric resource for future research on multimodal affect perception and emotionally expressive behavior in embodied agents.

The development of artificial agents, such as social robots and virtual characters, has highlighted the importance of emotional expression in human-facing interaction^{1,2}. For these agents to be accepted as credible social partners rather than mere tools, they must be able to convey their internal states and intentions in a manner that is both believable and readily interpretable by humans³. The failure to generate natural and coherent emotional expressions can lead to interaction breakdowns, reduce user trust^{4,5}, or plunge the agents into the “uncanny valley”⁶. Consequently, the field of affective computing is actively developing novel frameworks for generating expressive and contextually appropriate non-verbal behaviors for robots⁷.

Human emotional interaction is inherently multimodal^{1,8,9}, involving complex combinations of visual and auditory signals¹⁰. We do not perceive emotion from an isolated channel; rather, we seamlessly integrate cues from a speaker’s vocal prosody, facial expressions, body motion, and gestures to form a holistic impression¹¹. This principle of multimodal integration is well established in perception research, as illustrated by classic phenomena such as the McGurk effect in speech perception¹². Consequently, reliance on a single expressive modality may be insufficient in many human-agent interactions and can reduce perceived naturalness. Believability arises from the temporal and semantic coherence across multiple expressive channels^{13,14}.

At a foundational level, believable multimodal expressions demand both temporal synchronization^{15–17} and semantic coherence^{13,18}. However, the challenge of endowing agents with multimodal emotional expressivity goes beyond simply playing synchronized signals. The core difficulty lies in understanding the complex interplay and relative importance of different modalities^{19,20}. Prior work has shown that different modalities can contribute unequally to emotion judgments; for example, visual facial cues can outweigh vocal information in some settings^{21–23}. More broadly, multimodal emotion perception cannot be assumed to reflect a simple average of its component cues; instead, different modalities may be weighted unequally, and their combined percept may reflect dominance, attenuation, or limited context-dependent modulation^{24,25}. Such non-additive integration may influence perceived clarity, confidence, or intensity, but the form of this influence is likely to depend on the reliability and informativeness of the component cues. Yet, a comprehensive understanding of how such multimodal emotion judgments are weighted and integrated across channels remains incomplete.

¹State Key Laboratory of Mechanical System and Vibration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China. ²Shanghai Conservatory of Music, Shanghai 200031, China. ✉email: jhe@sjtu.edu.cn

Despite the substantial literature on facial and vocal cues, expressive body motion has received comparatively less attention²⁶. A key reason for this is that the emotional cues conveyed by body motion are often more subtle and implicit than those from more explicit channels like facial expressions or music²⁷. This subtlety poses a significant challenge for direct analysis. For many embodied artificial agents, especially robots with limited facial or vocal expressivity, body motion becomes a particularly important channel of non-verbal communication. This highlights the critical need for new methods to decode its emotional content. One way to address this challenge is to study body motion not in isolation, but in a naturalistic setting where it co-occurs with another emotionally informative signal. Such a setting makes it possible to examine how body motion contributes to emotion perception, whether independently or in relation to the accompanying signal, and whether that contribution varies across conditions. Musical performance provides a particularly suitable naturalistic context for this purpose. Music is widely recognized as a rich channel for emotional communication²⁸. More importantly, musical performance provides a naturalistic context where motion and music are intrinsically coupled. In this setting, sound-producing actions are directly linked to the acoustic output, while ancillary and expressive gestures remain functionally and temporally aligned with the musical structure. This combination means that performers' movements are not merely functional actions but also constitute a relevant source of visual information for observers²⁹. This intrinsic coupling makes musical performance a suitable naturalistic context for examining how auditory and visual cues are weighted and integrated during emotion perception, and this close relationship between sound and movement has already been leveraged in prior work to generate expressive movements for robots^{30,31}. However, despite this potential, it remains unclear how auditory and visual cues in musical performance are perceptually weighted, how the combined percept relates to its unimodal components, and whether the contribution of each modality remains stable across contexts.

Against this background, the present study investigates how music and body motion contribute to perceived valence and arousal in musical performance. Using EMOSIC, a synchronized dataset of audio and upper-body motion from performances on traditional Chinese plucked instruments, we compare audio-only, visual-only, and audiovisual judgments in order to characterize the relative contribution of the two modalities, the consistency with which each supports emotion judgments, and the overall pattern of audiovisual integration in this naturalistic context. The aim is to clarify how these two expressive channels relate to one another in perception. In addition to enabling this analysis, EMOSIC provides a non-Eurocentric resource for future research on multimodal emotion perception and emotionally expressive behavior in embodied agents.

Related work

To provide a foundation for this work, this section reviews prior research in three key areas: the expression of emotion in music, the communication of emotion through body motion, and the landscape of existing multimodal emotional datasets.

Emotion quantification in music and body motion

Emotional expression in both music and human motion is commonly studied through two primary frameworks: categorical and dimensional models³². Categorical models classify emotions into distinct categories such as happiness, sadness, and anger, which are intuitive and easily understood by the public^{33,34}. However, to capture the subtle nuances and continuous transitions inherent in emotional experiences, especially those evoked by music, dimensional models are often preferred in research. The most prominent of these is the circumplex model of affect, which represents emotions along the core dimensions of valence (from negative to positive) and arousal (from deactivation to activation)³⁵. This approach provides a more fine-grained model for quantifying the dynamic emotional trajectory within a musical piece.

Extensive research has established strong correlations between objective acoustic features and these emotional dimensions³⁶. Arousal is strongly linked to the temporal and energy-related characteristics of music. For instance, higher perceived arousal is consistently associated with a higher tempo³⁷, greater loudness³⁸ and wider pitch ranges³⁹. In contrast, valence is more closely related to melody, harmony, and timbre, with positive valence being associated with features such as large pitch variation³⁹, consonant harmonies³⁸ and bright timbre^{40–42}. These correlations between quantifiable acoustic features and perceived emotion highlight music's role as an effective channel for emotional communication.

In contrast to the relatively explicit acoustic cues in music, emotional expression through body motion presents additional challenges for computational analysis, particularly due to the lack of standardized metrics within music and emotion research. Body motion serves as an important channel for non-verbal emotional communication^{27,43}, and in many interactive contexts, emotional information conveyed through body motion is often perceived more rapidly than facial expressions⁴⁴. To overcome these challenges, a significant body of research has focused on identifying and quantifying a set of core kinematic features from high-dimensional, continuous motion data^{45,46}. These features can be broadly categorized into several domains. Energetic and temporal qualities, such as speed, acceleration, angular speed, and the overall quantity of motion, are fundamental⁴⁷. Spatially, emotional expression is conveyed through characteristics like limb contraction and expansion, overall body posture, head tilt, and the spatial volume occupied by the body, which can be measured by a 2D or 3D convex hull⁴⁸. Furthermore, the qualitative nature of movement, often described as its fluidity or smoothness, is quantitatively assessed through metrics like dimensionless jerk⁴⁹.

Numerous studies indicate that these kinematic features are intrinsically linked to the emotional dimensions of arousal and valence. Movement speed, in particular, is a frequently explored and highly informative parameter. A broad consensus in the literature suggests that faster movements are consistently associated with high-arousal emotions such as joy and anger, while slower movements are linked to sadness, fear, and neutrality^{50–53}. Similarly, high levels of acceleration and overall movement activity are characteristic expressions of anger and joy, whereas low levels are indicative of sadness^{54,55}. In addition, fluid and smooth movements are reliably associated with

positive emotions like happiness, whereas stiff, jerky, or low-fluidity motions are indicative of negative-valence emotions such as anger and fear⁵⁶.

Emotion and body motion in musical performance

Musical performance is fundamentally a multimodal phenomenon where auditory and visual channels are intrinsically linked^{29,57}. Beyond the functional movements required for sound production, musicians spontaneously employ expressive body motion to communicate structural and emotional intentions to the audience⁵⁸. Extensive research has established that these visual cues are not merely redundant byproducts but are powerful carriers of affective information^{59,60}. For instance, Castellano et al.⁶⁰ highlighted that the velocity of head movements in piano performance is a sensitive indicator of emotional arousal, while Chang et al.⁶¹ revealed that body sway in ensemble playing reflects joint emotional expression and correlates with perceived emotional intensity.

Building on these foundations, prior work has begun to explore how these modalities interact during emotion perception. Studies consistently show that while musicians' movements align with the emotional structure of the music^{62,63}, the integration of audio and visual cues by observers may involve intricate, non-linear interactions. Thompson et al.⁶² attempted to quantify these contributions using motion capture data from piano and violin performances, revealing a potential perceptual hierarchy: although observers can extract emotion-related information from visual cues alone, auditory features tend to exert a stronger influence on ratings when both modalities are present. These findings suggest that auditory and visual channels may be weighted differently by observers, a notion further supported by Moura et al.⁶⁴, who observed that the perceived "optimal" degree of body movement varies significantly with musical style and valence. More broadly, this line of work is consistent with audiovisual emotion research showing that observers do not treat auditory and visual affective cues as equally weighted, and that emotional congruency across modalities shapes perception^{19–21}. It is also compatible with work on the unity assumption, which argues that perceiving two channels as belonging together can modulate multisensory integration^{25,65}.

From a computational standpoint, such differential weighting across modalities has been formalized by reliability-weighted models of multisensory integration. Under the Maximum Likelihood Estimation (MLE) framework, the integrated percept is formed by weighting each channel in inverse proportion to its estimation variance, so that the more reliable modality exerts proportionally greater influence on the final percept^{66,67}. This principle has been shown to extend to audiovisual emotion perception in musical contexts: Petrini et al.⁶⁸ observed that in audiovisual presentations of musical improvisation, sound dominated the visual signal in emotion perception, and the relative weighting of the two channels shifted when cross-modal congruency was disrupted, consistent with a reliability-dependent integration strategy. Formal quantitative testing of MLE predictions typically requires parametric manipulation of cue reliability and psychometric estimation of unimodal variances⁶⁶. Whether the integration of audio and body motion cues in naturalistic performance settings follows a similar reliability-dependent pattern, and how such weighting is shaped by the intrinsic coupling between the two channels, has yet to be empirically established across diverse performance traditions.

Despite these significant insights, several avenues for further exploration remain. Many quantitative studies have focused on specific musical traditions (e.g., Western classical piano and violin), leaving room for investigation into whether these interaction patterns generalize to diverse cultural contexts and instrument families. Additionally, while the importance of cross-modal congruency is well-recognized, there is an opportunity to further refine our understanding of the specific modulatory role that body motion plays. Utilizing larger-scale and synchronized datasets from diverse performance traditions could provide a more granular view of this multimodal integration, potentially uncovering more nuanced patterns of how these expressive channels are integrated in naturalistic settings.

Multimodal emotional datasets

Empirical research in affective computing is fundamentally dependent on datasets, which provide a foundation for both understanding human expression and developing computational models. A persistent challenge in this domain is the reliable annotation of emotion, particularly for modalities where expressive cues are subtle. To address this, a common and effective strategy is to leverage multimodal data, where a more objective channel can be used to create a more robust ground truth. For instance, datasets like RECOLA⁶⁹, PPB-Emo⁷⁰ and DECAF⁷¹ augment participants' subjective emotion ratings with synchronously recorded physiological signals (e.g., ECG, MEG) to create a more objective and robust ground truth for analysis. This pairing of subjective ratings with objective physiological data is a powerful method for validating emotional labels.

This challenge of emotional annotation is particularly acute for expressive body motion. As discussed previously, the emotional content in movement is often holistic and difficult to label reliably without significant context. Following the established multimodal strategy, it is therefore logical to pair motion data with another channel to facilitate a more grounded investigation. Indeed, several multimodal datasets incorporating body motion have been developed. Some, like IEMOCAP⁷², FABO⁷³, and the recently proposed REED⁷⁴, combine motion with conversational speech or facial expressions. Others have focused on collecting expressive gestures, but often rely on actors performing predefined emotional scripts, such as in EMILYA^{75,76}. While valuable, the use of acted, non-naturalistic performances can limit the authenticity and generalizability of the findings. Furthermore, many existing resources derived from video, like EmoTV⁷⁷ and THEATER Corpus⁷⁸, can suffer from inaccuracies in 3D pose estimation and may lack the detailed kinematic data necessary for fine-grained analysis.

This review highlights the value of a dataset that pairs expressive body motion with a continuously varying, emotionally informative, and tightly coupled modality in a naturalistic setting. Within this context, musical performance offers a useful naturalistic paradigm for investigating how audio and body motion jointly contribute

to emotion perception: (1) **Intrinsic Coupling**: A performer's sound-producing actions directly generate the acoustic output, while ancillary gestures (e.g., body swaying or head movements) remain functionally and temporally aligned with the musical structure. Following the established framework that distinguishes functional from expressive gestures⁷⁹, musical performance provides a tightly coupled multimodal context for analysis. (2) **Naturalistic Context**: Musical performance provides a naturalistic setting for communicative emotional expression. While these expressions are often stylized or strategically shaped for performance purposes, and may not strictly correspond to the performer's immediate internal emotional state, they represent a genuine context for artistic communication that avoids the scripted nature of traditional acted datasets. (3) **Emotional Range**: Music can span a wide range within the Valence-Arousal space, providing a useful affective range for analysis. (4) **Analytical Clarity**: Expressive movements are superimposed upon a baseline of functional, sound-producing actions⁸⁰. This distinction between functional and expressive motion, while challenging to separate algorithmically, provides a useful theoretical framework for understanding how performers convey emotion beyond the mere act of playing the notes.

Therefore, to examine patterns of multimodal emotion integration and explore how these patterns generalize across cultures, a new type of resource is needed. Such a dataset must provide synchronized motion and audio data captured in an authentic setting. By featuring traditional Chinese instruments, EMOSIC aims to provide the affective computing community with a non-Eurocentric resource to facilitate more inclusive and systematic investigations into how expressive channels are integrated.

The EMOSIC dataset

To support the present investigation, we constructed EMOSIC, a synchronized dataset of audio and upper-body motion from traditional Chinese plucked-instrument performances. The dataset was designed to record naturally co-occurring musical audio and body motion in a synchronized format, enabling later analysis of their relationship in emotion perception. This section describes the data acquisition procedure, the performers and musical materials, the preprocessing pipeline, and an overview of the final dataset.

Data acquisition setup

The data were collected in a dedicated music performance and teaching classroom equipped with acoustic absorption panels, chosen to provide an environment that was both technically suitable and familiar to the performers. The acquisition setup included an 8-camera motion capture system (NOKOV, Mars 2H), a directional microphone for synchronized audio recording, and a reference video system (Fig. 1). Detailed hardware specifications, spatial configuration, and calibration procedures are provided in Supplementary Note 1. The motion capture system was used to record the 3D movements of the performers, while the audio system synchronously captured the musical sound. Together, these two streams form the core of the EMOSIC dataset, with the video system serving primarily as a reference for post-processing and data validation. The reference video was used exclusively for visual verification of the alignment markers and was not subjected to the same trimming or segmentation as the primary audio and motion capture streams.

Performers' motion was captured using a marker-based optical motion capture system. The setup comprised 8 high-speed infrared cameras arranged in a circular array around the performance area to provide stable multi-angle tracking coverage. Because many visually salient cues in seated plucked-instrument performance are concentrated in the upper body⁵⁹, we adopted a focused design centered on upper-body motion. This design prioritized the head, torso, and arms, which carry much of the observable movement variability in this performance context, while leaving lower-body motion outside the present capture scope. This focus supports a more targeted analysis of upper-body expressive cues in performance.

A custom set of 15 reflective markers was placed on the performer's upper body (Fig. 2). The placement protocol was designed to be both biomechanically informative and robust against occlusion from the instrument and performance movements. An 8-segment skeletal model was constructed from these markers, comprising the head/neck, the upper back, and three segments for each arm (shoulder, upper arm, and forearm). Reflecting the different degrees of freedom in performance, segments with a larger range of motion, such as the shoulders and arms, were defined by a triad of markers to robustly track both position and orientation. In contrast, segments with less complex motion, like the head/neck and upper back, were defined by dyads. This entire kinematic structure was captured at a sampling rate of 90 Hz. This rate is well suited to the present study's focus on macro-level upper-body gestures, such as head sway and arm trajectories, but it is not sufficient to resolve very rapid micro-movements such as finger vibrato or tremolo in detail. Accordingly, this study focuses on broader expressive movements rather than fine-motor instrumental techniques.

Audio was recorded in parallel using a cardioid directional microphone. The microphone was positioned to capture the instrument clearly enough for synchronization and low-level acoustic feature analysis while reducing incidental movement noise. Audio was recorded as a monophonic signal. To facilitate precise post-capture synchronization, a specific alignment protocol was executed at the start of each recording session. Instead of a traditional clapperboard, the performer executed a forceful, isolated string pluck. This action was designed to produce a distinct, high-amplitude onset in the audio waveform and a simultaneous, sharp kinematic peak in the motion data (the velocity peak of the right-hand marker), serving as a reliable multimodal synchronization marker.

Performers and musical material

To ensure the expressive quality of the performances, we recruited 10 professional musicians from a conservatory. The cohort included both professors and graduate students, each possessing over five years of professional training and extensive stage experience.

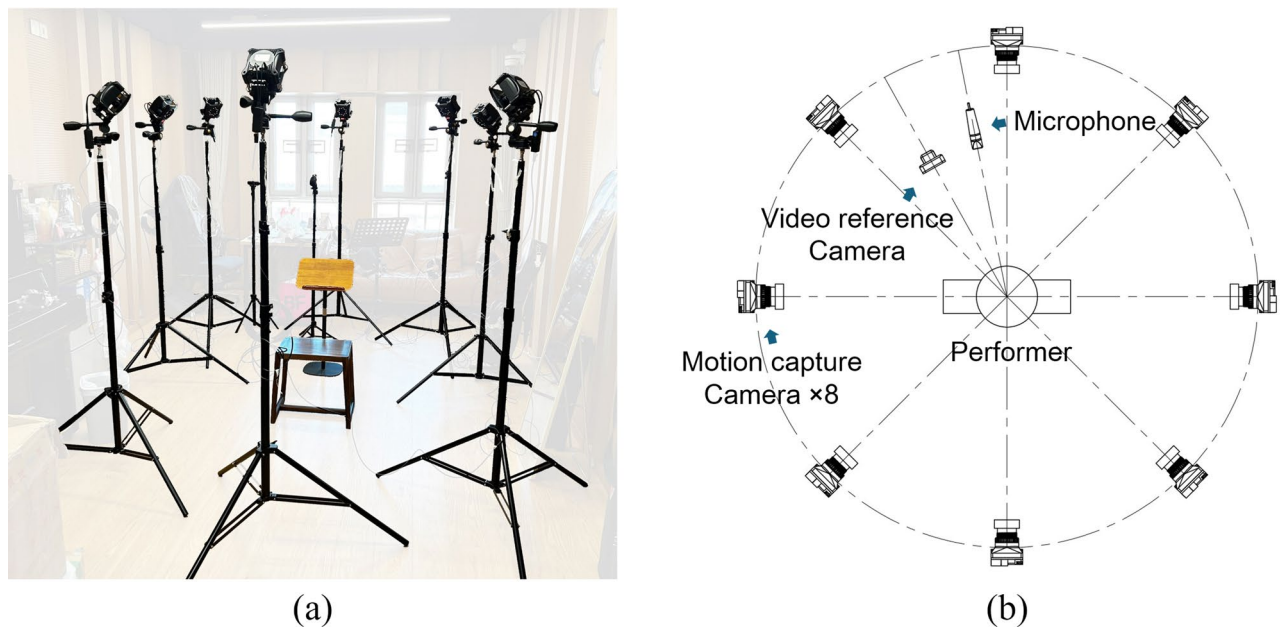


Fig. 1. Overview of the data acquisition setup. **(a)** Photograph of the data collection environment. **(b)** Top-down schematic diagram illustrating the layout of the key equipment, including the circular array of 8 motion capture cameras, the central performance area, the microphone, and the video reference camera.

The dataset features performances on a family of four traditional Chinese plucked instruments: the Zhong Ruan, Liuqin, Soprano Ruan, and Bass Ruan (Fig. 3). This instrumental family was chosen because it offers a relatively coherent performance context while still providing acoustic and gestural variation across related instruments. From a kinematic perspective, plucked-string instruments provide a useful performance context for studying upper-body motion. Performances on these instruments often involve visible upper-body movements beyond the minimum sound-producing actions. At the same time, the fundamental playing actions share a high degree of consistency across these four instruments. This kinematic commonality facilitates cross-instrument comparison by providing a more consistent functional baseline against which expressive movement variability can be examined. From an acoustic perspective, the multi-instrument design provides a broader range of timbral and pitch characteristics than a single-instrument dataset. The Zhong Ruan and Liuqin, as common lead instruments in Chinese orchestras, form the core of the dataset. To broaden the acoustic and pitch range of the dataset, we also included the highest- and lowest-pitched members of the Ruan family, namely the Soprano Ruan and Bass Ruan. Given that the Bass Ruan serves primarily as a harmonic and rhythmic foundation in traditional ensembles, its solo expressive repertoire is naturally more limited than that of lead instruments.

The repertoire spans several broad categories, including traditional and folk-derived works, contemporary and academic concert compositions, pedagogical exam materials, and a small number of internationally recognized pieces (see Supplementary Table S2 for the full piece-level list). The distribution of these repertoire types was not uniform across instruments, with Zhong Ruan and Liuqin contributing the broadest concert-style coverage, while Soprano Ruan and especially Bass Ruan were more limited in available solo material. Importantly, performers were not constrained to a single shared musical piece. Instead, each musician performed multiple pieces from their familiar repertoire, allowing stylistic variation while supporting fluent and natural performance. For each piece, the musicians provided a pre-performance annotation of the intended primary emotion; these labels are included as performer-intended reference information rather than as perceptual ground truth.

Data processing and segmentation

First, the raw motion capture data was inspected for occlusions and jitter. Gaps in marker trajectories (max lasting < 500 ms) were reconstructed using cubic spline interpolation. To reduce measurement noise while retaining the upper-body movement dynamics relevant to the present analysis, trajectories were smoothed using a fourth-order zero-phase Butterworth low-pass filter with a 25 Hz cutoff frequency. Subsequently, the audio and motion streams were aligned using a custom-developed Python script. This synchronization algorithm calculated the temporal offset by identifying the audio onset via amplitude thresholding and the kinematic peak of the pluck, which was defined as the maximum velocity of the right-hand marker. Based on this offset, the data streams were shifted onto a unified timeline. The alignment accuracy was manually verified against the reference video by inspecting the exact frame of string contact. The final synchronization precision was estimated to be within ± 2 frames, corresponding to approximately ± 22 ms at a 90 Hz sampling rate. Note that the reference video served primarily for visual validation of the alignment logic and was not included as a synchronized data stream in the final dataset.

Following alignment, the continuous recordings were segmented into manageable data units. We employed a sliding window to extract 8-second clips using a 50% overlap, which corresponds to a hop size of 4 s. This duration

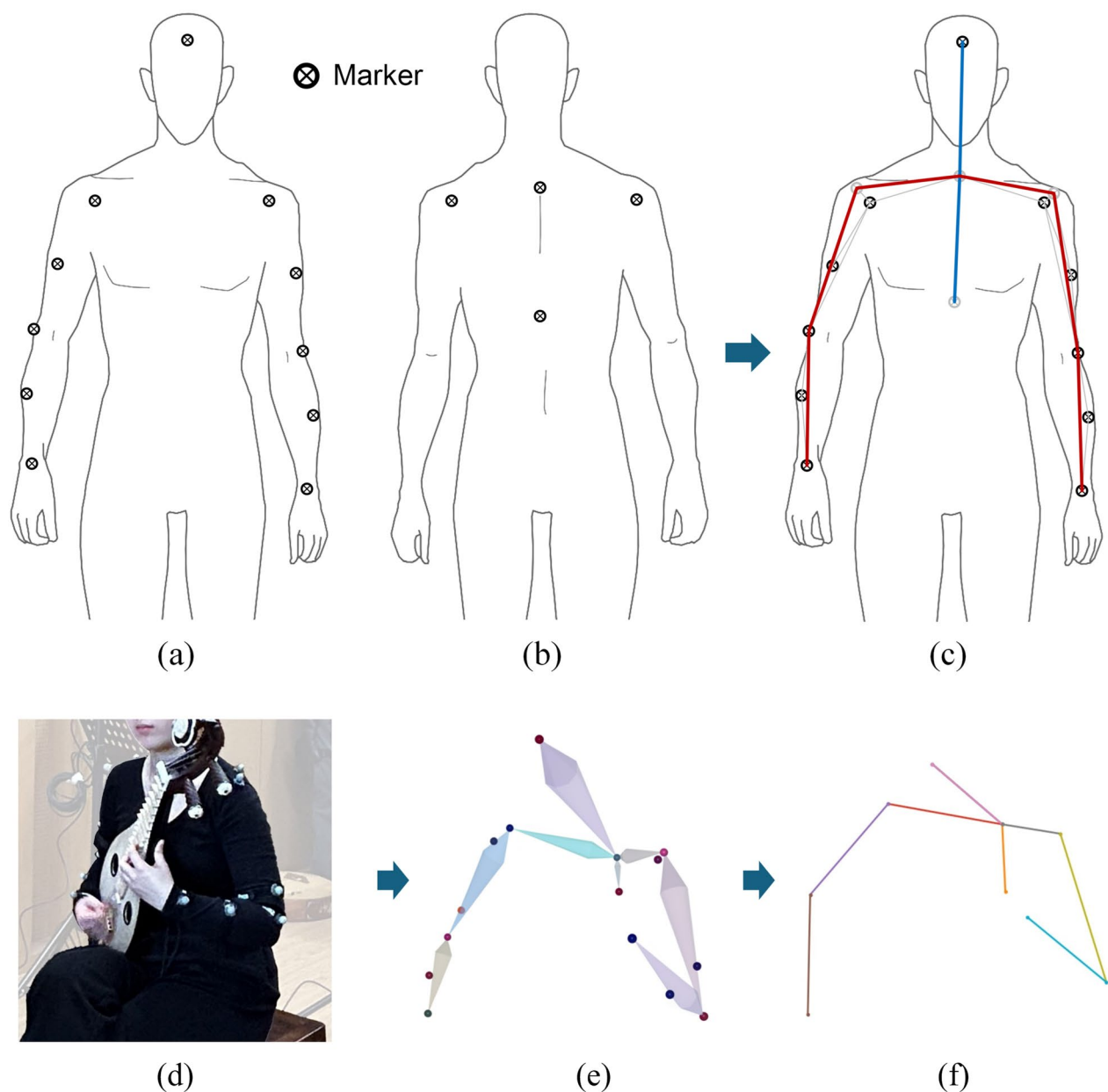


Fig. 2. Marker placement and skeletal model for upper-body motion capture. (a) Front view and (b) back view of the 15-marker set placement on key anatomical landmarks. (c) The resulting 8-segment skeletal model, with segments defined by marker triads (red) or dyads (blue) based on their rotational freedom. The data processing pipeline: (d) a performer wearing the marker set during a recording session; (e) the 3D marker data as captured by the system; and (f) an example frame from the final animation of the skeletal model.

was chosen as a balance between capturing short musical units and expressive gestures, and maintaining efficient annotation and downstream analysis. The overlapping strategy increased the number of analyzable clips while generating temporally shifted excerpts that, despite partial overlap, were centered on different musical moments. This process resulted in a total of 3,388 distinct clips.

The final dataset is provided in standard formats commonly used for motion and audio analysis. All motion capture clips are available in the BVH format, which includes both the skeletal hierarchy and the motion data sampled at 90 Hz. The corresponding audio for each clip is provided as a 16-bit, 16.0 kHz WAV file. This sampling rate was sufficient for the low-level acoustic features analyzed in the present study while keeping storage requirements manageable; however, it may limit applications that require higher-frequency timbral detail or production-quality music audio.

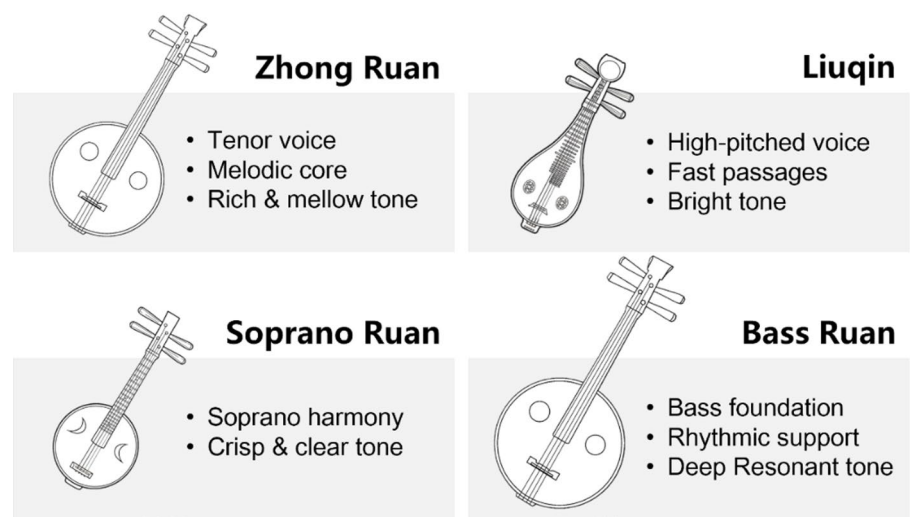


Fig. 3. The four traditional Chinese plucked instruments featured in the EMOSIC dataset: Zhong Ruan, Liuqin, Soprano Ruan, and Bass Ruan.

Metric	Value
Performers	
Number of Performers	10
Instruments	
Zhong Ruan	11,576 s (42.7%)
Liuqin	9,864 s (36.4%)
Soprano Ruan	5,416 s (20.0%)
Bass Ruan	248 s (0.9%)
Data Structure	
Total Number of Clips	3,388
Clip Duration	8 s
Total Duration	~ 7.5 h (27,104 s)
Technical Specifications	
Motion Capture Format	BVH, 90 Hz
Skeletal Model	8-Segment Upper-Body (15 markers)
Audio Format	WAV, 16.0 kHz, 16-bit

Table 1. Overview of the EMOSIC Dataset.

Dataset overview

The key statistics of the EMOSIC dataset are summarized in Table 1. As detailed in the table, the dataset comprises over 7.5 h of performance data from 10 professional musicians, segmented into 3,388 8-second clips. The distribution of data across the four instruments reflects the design strategy, with the Zhong Ruan and Liuqin forming the core of the collection. The relatively small share of the Bass Ruan (0.9%) is attributable to this scarcity of solo-performance material for bass-range instruments. To provide a descriptive overview of the dataset’s acoustic and kinematic variability, the following sections summarize several low-level feature distributions from both modalities. These descriptive analyses indicate substantial acoustic and kinematic variability relevant to affective analysis.

To illustrate the dataset’s acoustic diversity, Fig. 4 presents distributions of three canonical low-level features known to correlate with emotional dimensions. Following prior work in music emotion recognition³⁶, we selected Tempo and RMS Energy as representative descriptors related to arousal, and Spectral Centroid as a coarse descriptor of timbral brightness⁴¹ that has been associated in prior work with valence-related aspects of musical emotion^{36,40,42}. These distributions demonstrate a broad spectrum of musical characteristics relevant to affect. Tempo, a primary correlate of arousal measured in beats per minute³⁹, varies extensively from very slow passages below 40 BPM to extremely fast sections above 230 BPM, indicating substantial variation in pace across the dataset. Similarly, mean loudness demonstrates a wide dynamic range, spanning from quiet, delicate segments to forceful and intense passages. As a complementary timbral descriptor, we analyzed spectral centroid⁴⁰, which serves as a proxy for timbral brightness. Its wide range (660 Hz to 2810 Hz) reflects substantial timbral variation across the instrument family, from darker bass-range sonorities to brighter plucked attacks.

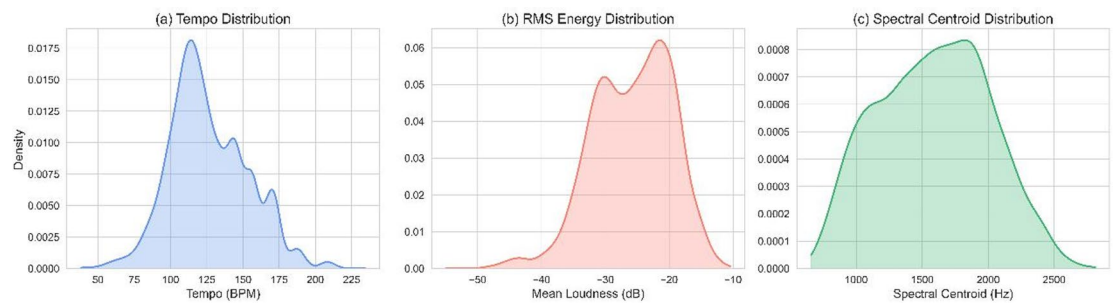


Fig. 4. Probability density distributions of key low-level audio features across all clips in the EMOSIC dataset. The panels show the distributions for (a) tempo, (b) mean RMS energy, and (c) mean spectral centroid.

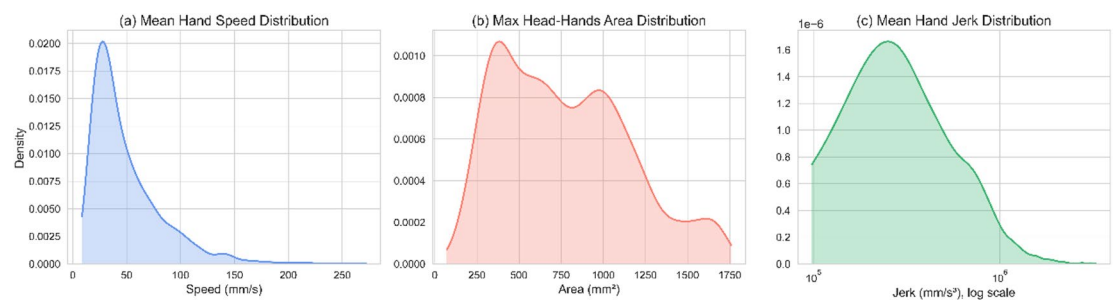


Fig. 5. Probability density distributions of key kinematic features across all clips in the EMOSIC dataset. The panels show the distributions for (a) mean hand speed, (b) max head-hands triangle area, and (c) mean hand jerk. The jerk distribution is shown on a logarithmic scale to better visualize its wide dynamic range, which spans several orders of magnitude.

Collectively, these distributions show that EMOSIC contains substantial acoustic variability relevant to affective analysis. However, they should be interpreted as descriptive evidence of feature diversity rather than as proof of balanced coverage of the full perceptual emotion space; the perceptual representativeness of the 60 selected stimuli is addressed separately in the experimental design section.

Similarly, to illustrate the kinematic diversity of the dataset, Fig. 5 presents distributions of three foundational kinematic features linked to emotional expression. Our selection was guided by comprehensive reviews of intelligible motion factors⁴⁵, which identify speed, spatial expansiveness, and jerk/fluidity as core dimensions of affective movement. The analysis reveals a wide spectrum of movement characteristics. The distribution of Mean Hand Speed, a commonly used indicator of arousal⁴⁷, shows that while most movements are performed at a moderate pace (peak at 27.9 mm/s), the dataset includes a long tail of more energetic gestures (max $\approx$ 272 mm/s). To characterize spatial expansiveness, which has been associated with affective movement in prior work⁴⁸, we computed the Max Head-Hands Triangle Area. Its distribution indicates that the dataset includes a broad range of postural expansiveness, from more contained to more expansive movements. Finally, the distribution of Mean Hand Jerk, a descriptor of movement fluidity that has been linked to affective movement quality, indicates variability from smoother to more abrupt motion patterns. Taken together, these analyses suggest that EMOSIC contains substantial variability in upper-body movement patterns relevant to affective analysis.

The preceding acoustic and kinematic analyses suggest that EMOSIC contains broad variability in features relevant to affective analysis, but they do not by themselves establish balanced coverage of perceptual emotion space. By providing synchronized audio and upper-body motion data from a naturally coupled performance context, EMOSIC offers a useful resource for studying multimodal expressive behavior in this repertoire. Beyond the present perception study, the dataset may also support analyses in which the auditory channel serves as contextual support for examining subtler patterns in body motion. Potential applications are discussed further in the Conclusion.

Experimental design and methodology

Participants

Thirty participants were recruited from the university campus and the local community through advertisements on campus BBS and social media. All participants reported normal or corrected-to-normal vision and normal hearing. The inclusion criteria for the study were an age range of 18–35 years and the ability to understand the experimental instructions. Exclusion criteria included any history of severe visual or auditory impairments, or known psychiatric or neurological conditions that could affect perceptual judgment. The study protocol was approved by the Institutional Review Board of Shanghai Jiao Tong University. All methods were performed in accordance with the relevant guidelines and regulations. Written informed consent was obtained from all

participants before the experiment, and participants received a nominal monetary compensation for their time. Additional consent for the publication of identifiable photographs was obtained where applicable. To reduce between-participant variability associated with cultural familiarity in this initial within-culture study, recruitment was limited to native Mandarin speakers who self-identified as ethnically Chinese. This design improves cultural consistency within the sample, but it also limits the immediate generalizability of the findings to culturally diverse or non-Chinese listener populations.

The sample size was calculated a priori using G*Power⁸¹. A power analysis for a repeated-measures ANOVA (within-factors) was conducted to detect a medium-sized effect (Cohen's $f=0.25$, corresponding to $\eta^2p=.06$) with a statistical power of 0.80 and an alpha level of 0.05 for three repeated measurements. The analysis indicated a required minimum sample size of 28 participants. To account for potential data loss or participant dropouts, we recruited a total of 30 individuals.

Stimuli sampling strategy

From the full dataset of 3,388 clips, a manageable and acoustically diverse subset of 60 clips was selected for the perceptual study. To obtain broad coverage in an acoustic feature space relevant to affect perception, we used a data-driven, two-stage stratified sampling strategy based on two coarse acoustic descriptors: tempo and spectral centroid.

The sampling procedure was implemented via a custom Python script. First, two acoustically grounded features were extracted from each of the 3,388 8-second audio clips using the librosa library: (1) Tempo, a measure of musical pace that is widely associated with arousal-related energy^{36,37}; and (2) Spectral Centroid, a coarse descriptor of timbral brightness⁴¹ that has been associated in prior work with valence-related aspects of musical emotion^{36,40,42}. These features were used as coarse descriptors to promote broad affect-relevant acoustic diversity in the selected stimuli. However, because perceptual valence in music is also shaped by other factors such as mode, harmony, and melodic contour, this sampling space should not be interpreted as guaranteeing uniform coverage of valence. Rather, the purpose of this procedure was pragmatic: to avoid clustering the selected clips within a narrow region of the acoustic feature space and to obtain a stimulus subset with broad affect-relevant variation for a manageable perceptual experiment. In this sense, the strategy should be understood as diversity-oriented rather than as a formal guarantee of exhaustive emotional-space coverage.

To stratify the dataset, the distribution of these two features across all clips was used to create a two-dimensional feature space. This space was divided into an 8×8 grid, resulting in 64 bins, with each bin representing a unique combination of tempo (from slow to fast) and spectral centroid (from dark to bright). This binning process ensured that clips with similar acoustic profiles were grouped together. This granularity was chosen to align closely with the target sample size of 60, allowing a near one-to-one correspondence between occupied bins and selected clips. The aim was to maximize coverage of the sampled acoustic space despite the discrete and non-uniform distribution of the tempo estimates (statistical coverage and mathematical derivation are detailed in Supplementary Note 3).

Finally, the two-stage sampling was executed. In the first stage, to maximize diversity, one clip was randomly sampled from each non-empty bin. This step guaranteed the inclusion of clips spanning a wide range of acoustic profiles. In the second stage, the remaining number of clips required to reach the target of 60 were randomly sampled from the entire pool of clips not selected in the first stage. This data-driven strategy reduced overrepresentation of densely populated acoustic regions in the dataset and yielded a stimulus set with broad acoustic variation. Because the sampling space was defined by tempo and spectral centroid, it should be interpreted as promoting broad coverage in affect-relevant acoustic feature space for the present perceptual comparison, without constituting a strict guarantee of balanced perceptual valence coverage. The distribution of the selected stimuli in the feature space is visualized in Fig. 6, illustrating broad coverage across the sampled acoustic range.

Experimental conditions

The perceptual experiment used a balanced incomplete-block design, with modality condition treated as a within-participant factor and stimulus-set assignment counterbalanced across participant groups. Each participant completed all three modality conditions, but did not rate the same performance clip more than once across conditions.

- Audio-Only (A-Only): Participants listened to the audio clips through headphones without any visual information.
- Visual-Only (V-Only): Participants viewed silent upper-body skeletal motion animations derived from the performer's motion-capture data.
- Audio-Visual (AV): Participants were presented with both the audio and the synchronized motion capture animation.

A key challenge in this design is to prevent participants from rating the exact same performance multiple times across different conditions, which could introduce memory and bias effects. The 60 selected stimuli were randomly partitioned into three distinct sets of 20 clips each (Set 1, Set 2, Set 3). The 30 participants were, in turn, randomly assigned to one of three groups (Group A, Group B, Group C; $n=10$ per group). Each participant completed 60 trials in total (20 per block).

The presentation of stimulus sets to participant groups under the different conditions was counterbalanced using a Latin Square design, as illustrated in Table 2. This structure ensures two key objectives: first, that each participant experiences all three modality conditions, and second, that across the entire experiment, each stimulus clip is rated by an equal number of participants ($n=10$) in each of the three conditions. This design

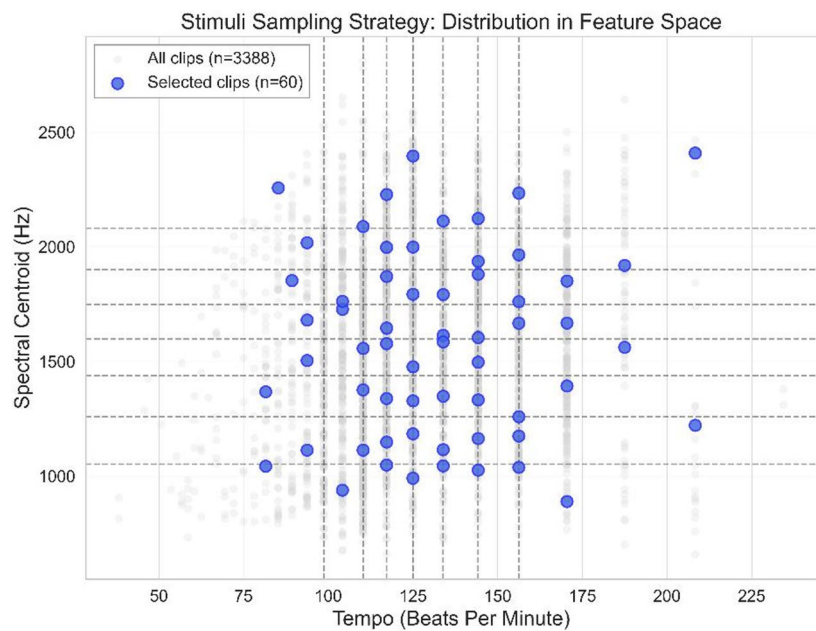


Fig. 6. Distribution of the stimuli sampling strategy in the feature space. Grey dots represent the full dataset ($n = 3,388$) and blue circles denote the 60 selected stimuli. Dashed lines indicate the 8×8 quantile grid used for stratified sampling to increase coverage across the acoustic feature space.

Participant Group	Audio-Only	Visual-Only	Audio-Visual
Group A ($n = 10$)	Set 1	Set 2	Set 3
Group B ($n = 10$)	Set 2	Set 3	Set 1
Group C ($n = 10$)	Set 3	Set 1	Set 2

Table 2. Assignment of stimulus sets to modality conditions across participant groups.

reduces stimulus-specific confounds by ensuring that each clip is evaluated in each modality condition across the full sample, while also counterbalancing the assignment of clip sets to conditions across participant groups. Within each block, trial order was randomized independently for each participant.

Procedure

The experiment was conducted individually for each participant in a quiet, sound-attenuated laboratory room with controlled lighting. As depicted in Fig. 7b, the experimental setup consisted of a 27-inch 2 K resolution monitor, a standard keyboard and mouse for interaction, and closed-back dynamic headphones for audio delivery. The entire experiment was administered through a custom-built web interface.

Upon arrival, participants provided written informed consent and completed a digital demographic questionnaire that included age, gender, education, and self-rated familiarity with traditional Chinese instrumental music. They then received standardized on-screen instructions explaining the concepts of valence and arousal, followed by a brief practice session to familiarize themselves with the task. For the main experiment, participants completed the three experimental blocks (A-Only, V-Only, AV) in a self-selected order to reduce subjective burden and fatigue.

All rating tasks were conducted using the primary annotation interface shown in Fig. 7a. Within each trial, an 8-second stimulus clip was presented once on the left panel of the interface. Immediately after, participants used the mouse to select a point on the two-dimensional V-A space on the right panel that best represented their perceived emotion. To ensure consistency with the circular structure of the circumplex model of affect, the selectable area was constrained within a unit circle. Descriptive anchor words (e.g., “Happy,” “Sad”) were displayed in the quadrants as orienting cues rather than as discrete response categories. After confirming their selection, the next trial began automatically. The entire session, including practice and a final screen for collecting feedback on task difficulty, lasted approximately 30 min per participant. Short breaks were allowed between blocks. All data, including participant ratings and demographic information, were linked to a unique, randomly generated user ID for anonymity and stored securely for offline analysis.

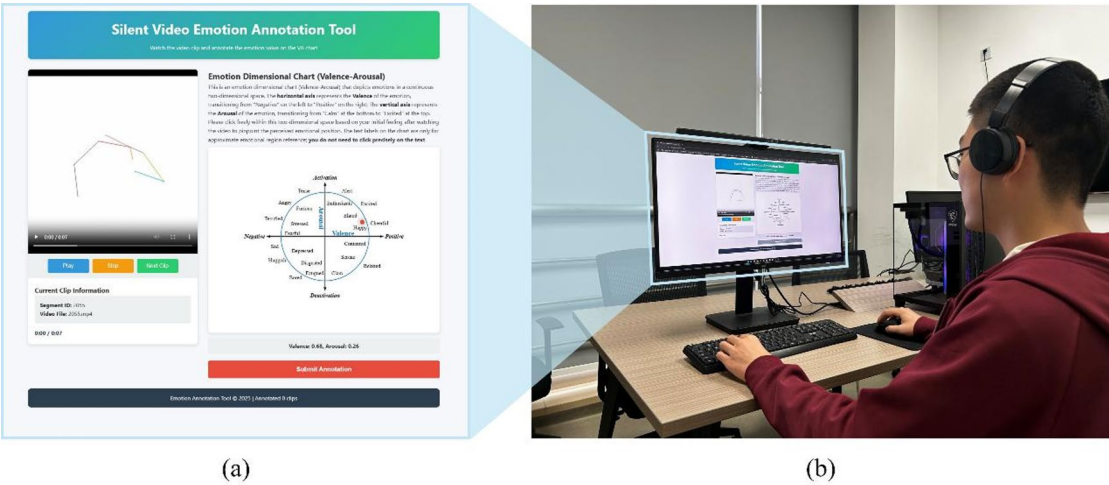


Fig. 7. Overview of the experimental setup and user interface. **(a)** A close-up of the annotation interface, featuring the media player on the left and the Valence-Arousal rating space on the right. **(b)** A participant engages in the rating task in the controlled laboratory environment.

Modality Condition	Valence-ICC	Arousal-ICC
Audio-Only	0.223	0.422
Visual-Only	0.018	0.315
Audio-Visual	0.246	0.493

Table 3. Inter-Rater Reliability (ICC2) for Valence and Arousal ratings across the three modality conditions.

Results
Participant demographics

Data were collected from 30 participants. Prior to the main analysis, the data were screened for completeness and obvious response anomalies. All participants completed the full set of trials, and no responses met the criteria for exclusion. The final sample consisted of 30 adults (17 male, 13 female) with an age range of 19 to 35 years ($M = 24.5$, $SD = 3.4$). All participants had at least a university-level education. In line with the study's focus on a specific cultural context, the majority of participants were from Mainland China, with a few from other regions in Asia, all of whom were of Chinese ethnicity and all native Mandarin speakers. Their self-reported musical familiarity was rated on a 5-point Likert scale, yielding a moderate average score ($M = 3.03$, $SD = 0.84$).

Inter-rater reliability

Before interpreting condition-level mean ratings, we assessed the inter-rater reliability (IRR) using the Intraclass Correlation Coefficient (ICC). Consistent with the balanced incomplete block design, ICC values were estimated separately for each of the three modality conditions. We report ICC estimates from a two-way random-effects model assessing the consistency of single raters (ICC2).

Table 3 summarizes the IRR results and highlights three main patterns. First, arousal ratings were consistently more reliable than valence ratings across all three conditions. Second, the ICC for Valence in the V-Only condition was exceptionally low ($ICC = 0.018$), indicating that participants could not form any meaningful consensus about the emotional valence based solely on the visual motion. Finally, the highest observed reliability was for arousal in the AV condition ($ICC = 0.493$), which falls within the moderate range. This pattern is consistent with clearer agreement in arousal judgments when both modalities were available, although differences between ICC values were not directly tested.

The near-zero agreement for valence in the V-Only condition warrants careful interpretation. Under the present visualization format, this result may reflect the limited availability of valence-relevant cues in skeletal body-motion displays presented without facial or situational context. Kinematic cues such as speed and movement range are often more informative for arousal than for valence, and similar movement patterns may remain compatible with multiple valence interpretations. Notably, the use of abstract skeletal animations was a deliberate methodological choice to isolate body kinematics from potential confounding variables such as facial affect, clothing, or performer identity. However, this high degree of abstraction may also have removed or attenuated micro-postural cues and perceived bodily tension that would otherwise support finer valence discrimination. Accordingly, the present findings point to a limitation of sparse skeletal representations for valence decoding in this paradigm, rather than demonstrating that body motion in general cannot convey valence. Future work should compare more ecologically valid visualization formats to disentangle limitations of embodiment format from limitations of kinematic information itself.

Consequently, all subsequent analyses involving V-Only valence means should be interpreted with substantial caution and are treated as tentative rather than strong evidence of stable visual valence decoding.

Distribution of perceptual ratings

To visualize the overall distribution of the collected emotional ratings, we generated two-dimensional kernel density estimates on the Valence-Arousal (V-A) space for each of the three experimental conditions. As shown in Fig. 8, these heatmaps highlight regions of higher rating density and provide a qualitative overview of the affective space sampled by the stimulus set. Across all three conditions, the ratings were not concentrated around the neutral origin but were distributed more broadly across non-neutral regions of the V-A space. This pattern suggests that the selected stimuli elicited affective judgments that were generally non-neutral across conditions.

Clear descriptive differences are nevertheless visible in the shape and structure of the distributions across the three conditions. The A-Only (Fig. 8a) and AV (Fig. 8c) conditions show broadly similar distributional structures, with density peaks appearing across multiple quadrants of the V-A space. The descriptive similarity between these two heatmaps is consistent with an audio-dominant pattern, in which the audiovisual percept more closely resembles the audio-only percept than the visual-only percept. By contrast, the distribution for the V-Only condition (Fig. 8b) shows a visibly different pattern. Although regions of higher density are still visible, these clusters appear less sharply defined. The V-Only distribution is relatively more diffuse, especially along the valence dimension, which visually accords with the near-zero inter-rater agreement observed for V-Only valence in the previous section.

Taken together, these distributional patterns provide qualitative context for the quantitative analyses that follow. They are visually consistent with a stronger influence of the audio channel and with the relative ambiguity of the visual-only condition, but they should be interpreted as descriptive rather than inferential evidence.

Cross-modal correspondence

One question of interest in this study is whether the auditory and visual channels in musical performance show corresponding patterns of perceived emotion across clips. To examine this question, we calculated Pearson correlation coefficients between the mean participant ratings for the A-Only and V-Only conditions across the 60 stimulus clips. This analysis assesses whether clips rated as relatively higher or lower in the audio-only condition tended to receive correspondingly higher or lower ratings in the visual-only condition, separately for arousal and valence.

The scatter plots in Fig. 9 indicate a clear cross-modal correspondence for arousal and a weaker, more tentative correspondence for valence. For arousal, we found a significant strong positive correlation between the A-Only and V-Only mean ratings ($r = .754, p < .001$). As shown in Fig. 9b, clips judged as more arousing in the audio-only condition also tended to be judged as more arousing in the visual-only condition. For valence, a significant moderate positive correlation was also observed ($r = .402, p = .001$), although this result must be interpreted in light of the very low V-Only valence reliability reported above. As shown in Fig. 9a, this relationship suggests a weak population-level tendency for clips rated as more positive or negative in the audio-only condition to be rated in the same general direction in the visual-only condition.

Beyond the correlation, the regression slopes provide additional descriptive information about the relative dynamic range of the two conditions. For arousal, the regression slope was 0.653, while for valence, it was 0.225. Both slopes being below 1 suggests that the range of ratings in the V-Only condition was more compressed than in the A-Only condition. Numerically, this compression was stronger for valence than for arousal, although the valence result should be interpreted with substantial caution. This descriptive pattern is consistent with the reliability analysis in suggesting that the visual-only condition conveyed a weaker and more compressed emotional signal than the audio-only condition, especially for valence under the present skeletal-display format.

Taken together, these results indicate a clearer cross-modal correspondence for arousal than for valence. The positive arousal correlation indicates that clips receiving higher or lower arousal ratings in the A-Only condition tended to receive correspondingly higher or lower arousal ratings in the V-Only condition. A positive correlation was also observed for valence, but this relationship should be interpreted more cautiously because agreement

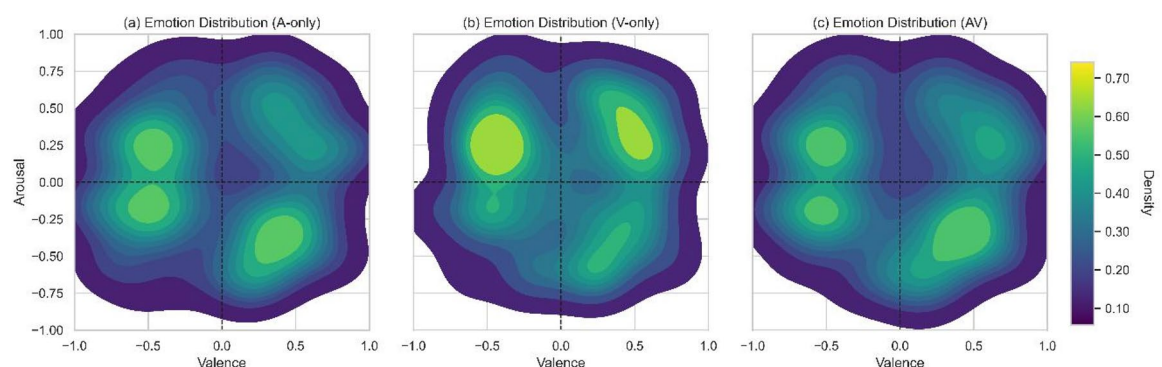


Fig. 8. Kernel density estimate heatmaps of emotion ratings on the Valence-Arousal space for the (a) Audio-Only, (b) Visual-Only, and (c) Audio-Visual conditions. Brighter colors indicate a higher density of ratings.

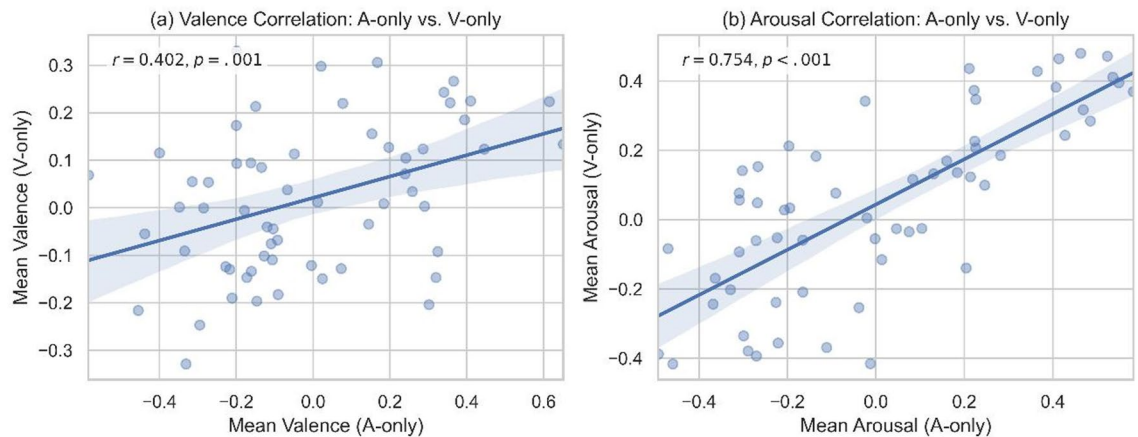


Fig. 9. Correlation between mean emotion ratings in the Audio-Only and Visual-Only conditions. Scatter plots for (a) Valence and (b) Arousal across the 60 stimulus clips. The shaded areas represent the 95% confidence interval of the regression fit.

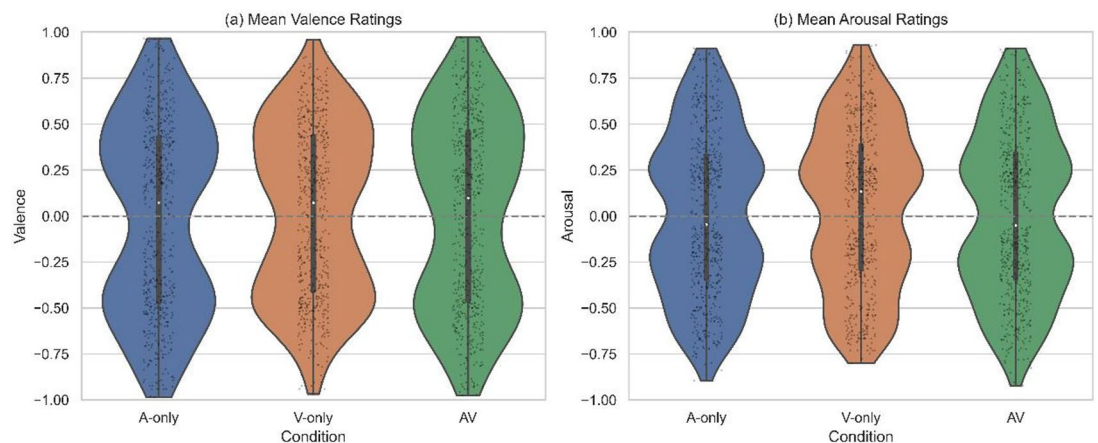


Fig. 10. Violin plots of (a) Valence and (b) Arousal ratings across the three modality conditions. The internal boxplots indicate the median and interquartile range; individual dots represent the raw ratings provided by all participants across all stimuli.

on valence in the V-Only condition was extremely low. Overall, the correlation analysis suggests that audio and body motion shared some affect-related structure, especially for arousal, while the valence relationship appeared weaker and less stable.

Aggregate-level comparisons across modality conditions

To examine whether emotion judgments differed across modality conditions, we compared ratings across the three conditions using a one-way repeated-measures analysis of variance (ANOVA). In addition to mean valence and arousal ratings, we also examined two derived measures, Emotional Intensity and Rating Magnitude, as descriptive summaries of how far ratings lay from the neutral point in the two-dimensional space and from the midpoint of each single dimension, respectively.

Prior to the main analysis, we assessed the statistical assumptions for the repeated measures ANOVA. Shapiro-Wilk tests on the model residuals indicated deviations from normality ($p < .001$), a common characteristic of bounded rating scales. However, given the sample size, the balanced repeated-measures design, and the relatively mild distributional deviations⁸², we retained the parametric analysis as the primary analysis. To assess robustness, we also conducted non-parametric Friedman tests for all main analyses. The non-parametric analyses generally supported the overall pattern of results, but the effect on Emotional Intensity did not remain significant in the Friedman test, indicating greater sensitivity to distributional assumptions (see Supplementary Note 4).

We first tested whether combining modalities shifted the average perceived valence or arousal. The ANOVA, however, showed no significant main effect of condition on the mean valence ratings ($F(2, 58) = 0.65, p = .524, \eta^2 p = .022$) and no significant main effect on the mean arousal ratings ($F(2, 58) = 1.15, p = .323, \eta^2 p = .038$). As visualized in the violin plots and overlaid raw ratings in Fig. 10, both the overall distributions and the mean ratings for valence and arousal were broadly similar across the three conditions. These null effects provide no

evidence that the audiovisual condition systematically shifted the perceived emotional location in the V-A space relative to the unimodal conditions.

Because mean valence and arousal did not differ across conditions, we next examined whether the conditions differed on additional derived measures of rating geometry. Specifically, we asked whether modality differences might appear not in the mean location of ratings, but in how far ratings were distributed from the neutral point in the V-A space or from the midpoint of each single dimension. To describe these additional aspects of the rating structure, we computed two derived metrics. First, we defined Emotional Intensity as the Euclidean distance from the neutral origin in the V-A space, using this quantity as an aggregate index of overall perceived emotional strength. In the present analysis, larger values on this metric indicate ratings located farther from the neutral point in the two-dimensional affective space, and were interpreted as reflecting stronger overall emotional responses. Second, we computed Rating Magnitude as the absolute value of each rating, in order to characterize how far judgments on each dimension departed from the neutral midpoint. In the present study, ratings farther from the neutral midpoint were taken to reflect more pronounced or more decisive judgments on that dimension, whereas ratings closer to zero were interpreted as relatively more ambiguous or less differentiated.

$$\text{Emotional Intensity: } I = \sqrt{V^2 + A^2}$$

$$\text{Valence Rating Magnitude: } M_V = |V|$$

$$\text{Arousal Rating Magnitude: } M_A = |A|$$

where V and A denote valence and arousal coordinates, respectively.

A repeated measures ANOVA on Emotional Intensity revealed a main effect of condition ($F(2, 58) = 3.98, p = .024, \eta^2 p = .121$), suggesting a difference across modalities in this aggregate index of overall perceived emotional strength. However, the corresponding Friedman test did not reach significance ($\chi^2(2) = 1.87, p = .393$), indicating that the Emotional Intensity effect was more sensitive to distributional assumptions than the other aggregate-level findings reported here. Bonferroni-corrected post-hoc comparisons showed that Emotional Intensity was significantly higher in the A-Only condition than in the V-Only condition ($p\text{-corr} = 0.042$). This contrast suggests that ratings in the A-Only condition were displaced farther from neutrality than those in the V-Only condition, consistent with a stronger contribution of the auditory channel to overall perceived emotional strength. The AV condition also showed numerically higher Emotional Intensity than V-Only, but this difference did not remain significant after Bonferroni correction ($p\text{-corr} = 0.080$). Taken together, these results suggest a stronger contribution of the auditory channel to overall Emotional Intensity, with no reliable evidence that visual information further increased it.

A separate repeated measures ANOVA on Valence Rating Magnitude also yielded a significant main effect of condition ($F(2, 58) = 6.61, p = .003, \eta^2 p = .186$). This condition effect was also significant in the Friedman test ($\chi^2(2) = 8.47, p = .015$), indicating that differences in Valence Rating Magnitude across conditions were relatively robust to distributional assumptions. Bonferroni-corrected post-hoc tests identified two significant pairwise differences. Valence Rating Magnitude was significantly greater in the A-Only condition than in the V-Only condition ($p\text{-corr} = 0.020$), indicating that valence judgments in the audio-only condition were farther from the neutral midpoint. Valence Rating Magnitude in the AV condition was also significantly greater than in the V-Only condition ($p\text{-corr} = 0.016$). As shown in Fig. 11(b), this pattern suggests that valence judgments in the V-Only condition remained closer to neutrality, whereas both A-Only and AV yielded more pronounced valence ratings. In the present analysis, this was interpreted as greater decisiveness of valence judgment in the conditions containing auditory information.

Taken together, the aggregate-level analyses indicate an asymmetrical pattern across modality conditions. Mean valence and arousal ratings remained broadly similar across the three conditions, and no condition effect was observed for Arousal Rating Magnitude ($F(2, 58) = 0.11, p = .896$). The clearest aggregate-level difference emerged in Valence Rating Magnitude, where ratings in the V-Only condition were closer to the neutral

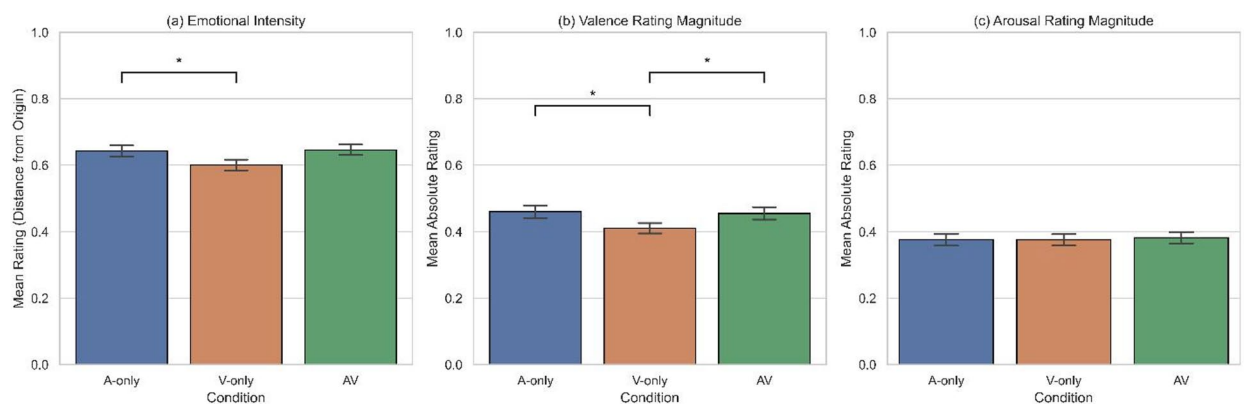


Fig. 11. Bar plots of (a) mean Emotional Intensity, (b) mean Valence Rating Magnitude, and (c) mean Arousal Rating Magnitude across the three modality conditions. Error bars represent 95% confidence intervals. Asterisks denote statistically significant pairwise differences (* $p < .05$).

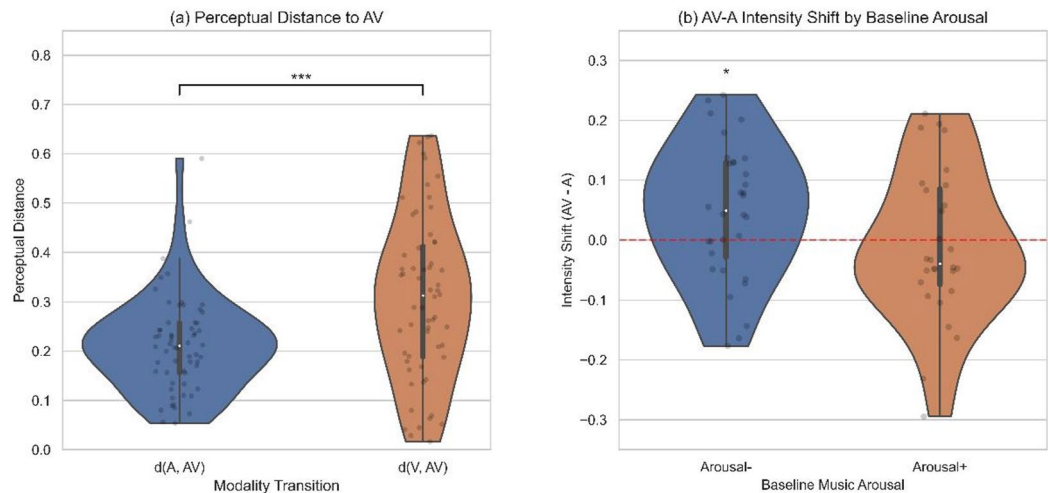


Fig. 12. Violin plots of (a) perceptual distances from the integrated AV percept to constituent modalities, and (b) intensity shifts (AV minus A) stratified by baseline music arousal. The internal boxplots indicate the median and interquartile range; individual dots represent the calculated values for each of the 60 stimulus clips. The red dashed line in (b) represents zero shift. Asterisks denote statistically significant differences (*** $p < .001$, * $p < .05$).

midpoint, whereas both A-Only and AV yielded more pronounced valence judgments. In the present analysis, this pattern was interpreted as showing that auditory information supported clearer valence judgments than visual-only presentation. At the same time, because the AV condition was not shown to exceed the A-Only condition, these results do not by themselves demonstrate that visual information reliably increased valence beyond the auditory baseline. In addition, the condition effect on Emotional Intensity was less stable across statistical assumptions, and is therefore interpreted more cautiously.

These aggregate-level findings suggest that any contribution of visual information is likely to be limited and may depend on stimulus-specific characteristics. To examine this possibility, we next turned to an exploratory clip-level analysis of transitions across conditions.

Exploratory clip-level analysis of cross-modal transitions

While the comparison of group-level means reveals overall trends, it may obscure the stimulus-wise dynamics of how individual clips are perceived across modalities. To explore this possibility, we conducted an exploratory clip-level analysis based on the mean ratings of the 60 stimulus clips. We defined two descriptive metrics to characterize transitions between conditions in the Valence-Arousal space: (1) **Perceptual Distance**, defined as the Euclidean distance between modality conditions (e.g., $d(A, AV) = \sqrt{(V_A - V_{AV})^2 + (A_A - A_{AV})^2}$) for each clip, and (2) **Intensity Shift**, defined as the change in radial distance from the origin between two conditions (e.g., $I_{AV} - I_A$).

We first compared whether the AV percept was closer to the A-Only or the V-Only condition in the V-A space by examining $d(A, AV)$ and $d(V, AV)$. As illustrated in Fig. 12(a), the distribution of $d(A, AV)$ ($M = 0.21$, $SD = 0.10$) is significantly lower than that of $d(V, AV)$ ($M = 0.31$, $SD = 0.17$) ($t(59) = -4.13$, $p < .001$, $d = 0.53$). Because both dimensions were rated on the same bounded scale (a unit circle in V-A space), standardization of the two axes prior to distance computation was not applied. This pattern suggests that, across the 60 stimulus clips, the AV ratings were closer to the A-Only ratings than to the V-Only ratings. Accordingly, this distance pattern is more consistent with a stronger contribution from the auditory condition.

We then defined Intensity Shift with respect to the A-Only condition (i.e., $I_{AV} - I_A$). This choice was motivated by the distance comparison above, which indicated that the AV ratings tended to be closer to A-Only than to V-Only, and by the lower stability of the V-Only ratings, especially for valence. Under this definition, Intensity Shift captures the extent to which adding visual information changed the radial distance of ratings from the neutral origin once an auditory reference was already present.

The global mean shift was positive but not statistically significant ($M = +0.02$, $p = .198$), providing no evidence for a consistent increase in radial distance from A-Only to AV across all stimuli. This motivated a closer exploratory look at whether the AV-A difference varied as a function of stimulus properties. We stratified the 60 clips into a Low-Arousal subgroup (Arousal -, $n = 32$; clips with negative A-Only mean arousal) and a High-Arousal subgroup (Arousal +, $n = 28$; clips with non-negative A-Only mean arousal) to examine whether the AV-A Intensity Shift varied with baseline auditory arousal.

The subgroup analysis, visualized in Fig. 12(b), suggested different patterns across the two subsets. For the Low-Arousal group, the AV condition showed a significant positive Intensity Shift relative to A-Only ($M = +0.05$, $t(31) = 2.36$, $p = .025$). This pattern is consistent with the possibility that visual information was associated with a larger radial displacement from the neutral origin when baseline auditory arousal was relatively low. By contrast,

no significant Intensity Shift was observed in the High-Arousal group ($M = -0.01$, $t(27) = -0.43$, $p = .670$). One possible interpretation is that visual information contributed less when baseline auditory arousal was already high, leaving less room for additional radial displacement from the neutral origin. However, given the post hoc nature of this subgrouping, this account should be regarded as speculative and hypothesis-generating.

Collectively, these exploratory clip-level analyses suggest that the audiovisual ratings remained closely aligned with the auditory condition, while positive Intensity Shifts appeared only in a subset of lower-arousal clips and did not characterize the full stimulus set. These interpretations are based on clip-level mean ratings in Valence-Arousal space and should therefore be regarded as descriptive and exploratory, rather than as evidence for a specific underlying integration mechanism.

Conclusion and future work

This paper investigated how music and body motion contribute to emotion perception in musical performance through a controlled perceptual study supported by EMOSIC, a synchronized multimodal dataset of traditional Chinese plucked-instrument performances. Our findings showed clearer cross-modal correspondence for arousal than for valence and indicated that body motion alone yielded weaker and less reliable judgments than the audio-only and audiovisual conditions. The results also revealed an asymmetrical pattern of integration: emotion judgments in the audiovisual condition were closer to the audio-only condition than to the visual-only condition, consistent with an audio-dominant perceptual pattern. At the same time, audiovisual presentation did not significantly shift mean valence or arousal, and the condition effect on emotional intensity was sensitive to statistical assumptions. Within this overall pattern, the clearest robust effect was that valence judgments in the visual-only condition were closest to neutrality, indicating that body motion alone provided a comparatively ambiguous basis for valence evaluation. Exploratory clip-level analyses, however, pointed to a more nuanced possibility: when baseline auditory arousal was lower, adding body motion was associated with a modest increase in perceived emotional intensity, consistent with a limited context-dependent contribution of visual motion.

These findings suggest an asymmetric form of multimodal integration in which the two channels make unequal contributions to the final percept. The observed pattern is qualitatively consistent with reliability-weighted accounts of multisensory integration^{66,67}, which predict that less informative channels receive proportionally lower perceptual weight; here, the near-zero agreement for visual-only valence and the close correspondence between audiovisual and audio-only judgments accord with this prediction. In this musical-performance context, the auditory channel appears to provide the primary perceptual anchor, while body motion contributes a secondary and more constrained source of affective information. Its role in audiovisual perception therefore appears limited but non-negligible: it does not substantially relocate the perceived emotion in valence-arousal space, yet it may provide a supplementary contribution to the final judgment under certain conditions. The exploratory low-arousal pattern further suggests that this contribution may become more noticeable when baseline auditory arousal is lower. For human-robot interaction and related affective computing applications, these findings suggest that coordinating body motion with a primary auditory channel may be a useful direction for improving cross-modal coherence and the readability of emotional expression in embodied agents.

Beyond the present perception study, EMOSIC provides a useful synchronized audio-motion resource for research on multimodal affective expression in musical performance, while also contributing a non-Eurocentric empirical resource for related research communities. Its value lies not only in the temporal alignment between audio and motion, but also in its naturalistic performance context, professional performers, more than 7.5 h of recordings, and the broad acoustic and kinematic variability documented in the dataset. These characteristics make it suitable for future work on multimodal affect modeling, emotion recognition, and analyses of how auditory and movement cues relate to one another in musical performance. The dataset may also serve as a naturalistic benchmark for motion generation studies and related research on embodied agents, including work that draws on expressive movement patterns from musical performance to inform research on emotionally expressive behavior in artificial agents³¹.

Several limitations should be considered when interpreting the present findings. First, both the participant sample and the stimulus materials were culturally homogeneous: the study was conducted with Chinese listeners using performances drawn from traditional Chinese plucked-string repertoire, and familiarity with this repertoire may have influenced affective judgments, although this factor was not directly modeled in the present analyses. An exploratory examination was considered but not pursued, as the limited variance in self-reported familiarity scores ($M = 3.03$, $SD = 0.84$ on a 5-point scale) and the modest sample size ($N = 30$) constrained the statistical power and interpretability of such an analysis. Future work should therefore examine whether the present pattern generalizes across culturally diverse listener groups and should test more directly how familiarity with the musical tradition modulates rating behavior. Second, the visual stimuli consisted of abstract skeletal animations rather than naturalistic visual renderings of human performers, so the present findings concern the perception of upper-body kinematics in an abstracted display format rather than the full visual experience of live or video-recorded performance. This limitation is especially important for valence, as the near-zero inter-rater reliability in the visual-only condition means that all quantitative interpretations that rely on visual-only valence means should be treated with caution. Finally, the low-arousal facilitation pattern emerged from exploratory clip-level analyses and should therefore be treated as hypothesis-generating; future studies should test this pattern in larger, independently designed confirmatory studies. In addition, extending the present descriptive approach through parametric manipulation of cue reliability would enable formal testing of whether emotion integration in this naturalistic performance context conforms to the quantitative predictions of reliability-weighted models.

In summary, the present findings indicate that emotion perception in this musical-performance context was primarily audio-dominant, while body motion alone remained a weaker and more ambiguous source of affective information, especially for valence. Any additional visual contribution appeared limited and context-dependent, although the exploratory analyses suggested that body motion may provide a modest facilitatory contribution

to perceived emotional intensity when baseline auditory arousal is lower. The EMOSIC dataset provides a non-Eurocentric empirical resource for future research on multimodal affect perception, and the present findings may also offer a useful perspective for thinking about emotionally expressive multimodal behavior in embodied agents.

Data availability

The EMOSIC dataset generated and analysed during the current study is publicly available in the Zenodo repository, <https://doi.org/10.5281/zenodo.17623118>.

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This study protocol was approved by the Institutional Review Board of Shanghai Jiao Tong University. The authors note that the primary affiliation of Biyun Zhang is Shanghai Conservatory of Music.

Author contributions

K.M. designed the methodology, conducted the experiments, performed the data processing and analysis, and wrote the main manuscript. B.Z. was responsible for the musical aspects of the study, including the design of performance tasks, and assisted in revising the manuscript. J.H. supervised the project, assisted in the overall research design, and contributed to the final version of the manuscript. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to J.H.

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